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# Activity-1

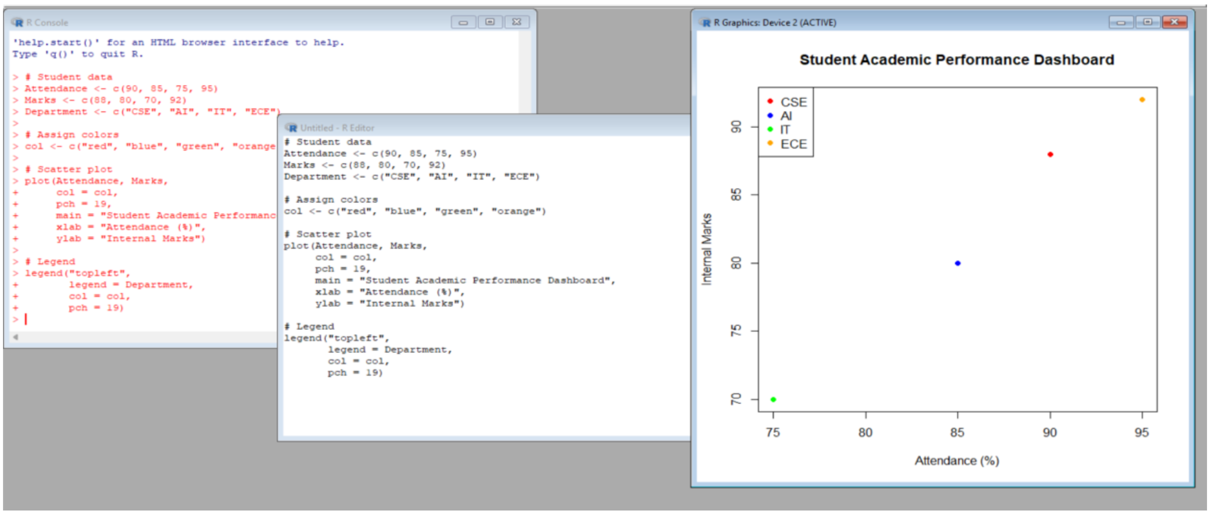
# R-Programming

## Scenario 1: Student Academic Performance Dashboard

### A. Code

```
Attendance <- c(90, 85, 75, 95)

Marks <- c(88, 80, 70, 92)

Department <- c("CSE", "AI", "IT", "ECE")

col <- c("red", "blue", "green", "orange")

plot(Attendance, Marks,

    col = col,

    pch = 19,

    main = "Student Academic Performance Dashboard",

    xlab = "Attendance (%)",

    ylab = "Internal Marks")

legend("topleft",

      legend = Department,

      col = col,

      pch = 19)
```

**Output:**

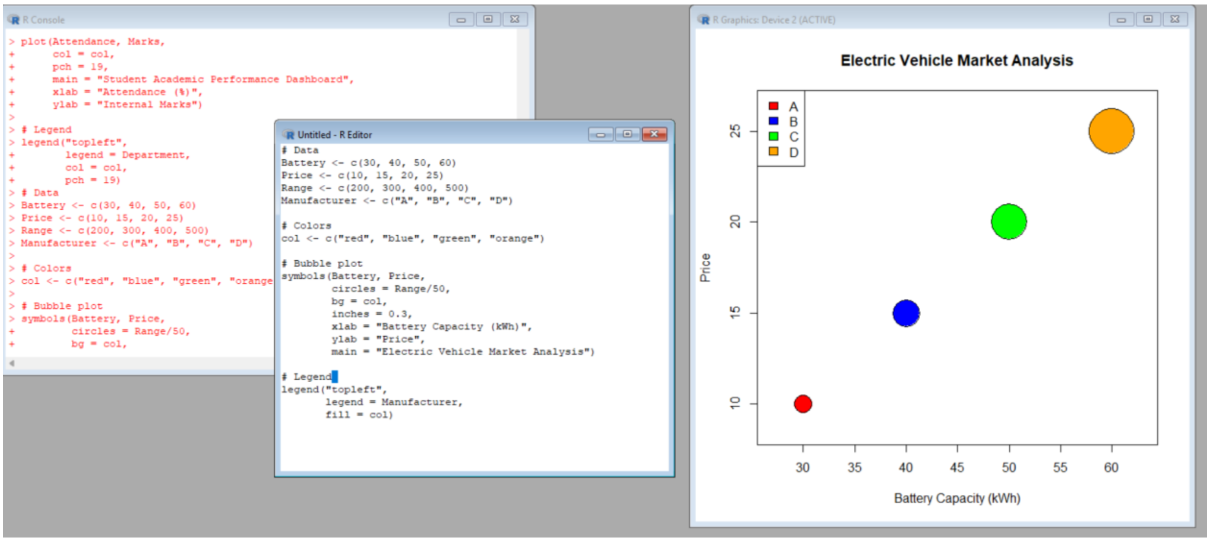


**Scenario 2: Electric Vehicle Market Analysis**

**A. Code**

```
Battery <- c(30, 40, 50, 60)
Price <- c(10, 15, 20, 25)
Range <- c(200, 300, 400, 500)
Manufacturer <- c("A", "B", "C", "D")
col <- c("red", "blue", "green", "orange")
symbols(Battery, Price,
        circles = Range/50,
        bg = col,
        inches = 0.3,
        xlab = "Battery Capacity (kWh)",
        ylab = "Price",
        main = "Electric Vehicle Market Analysis")
legend("topleft",
       legend = Manufacturer,
       fill = col)
```

**Output:**

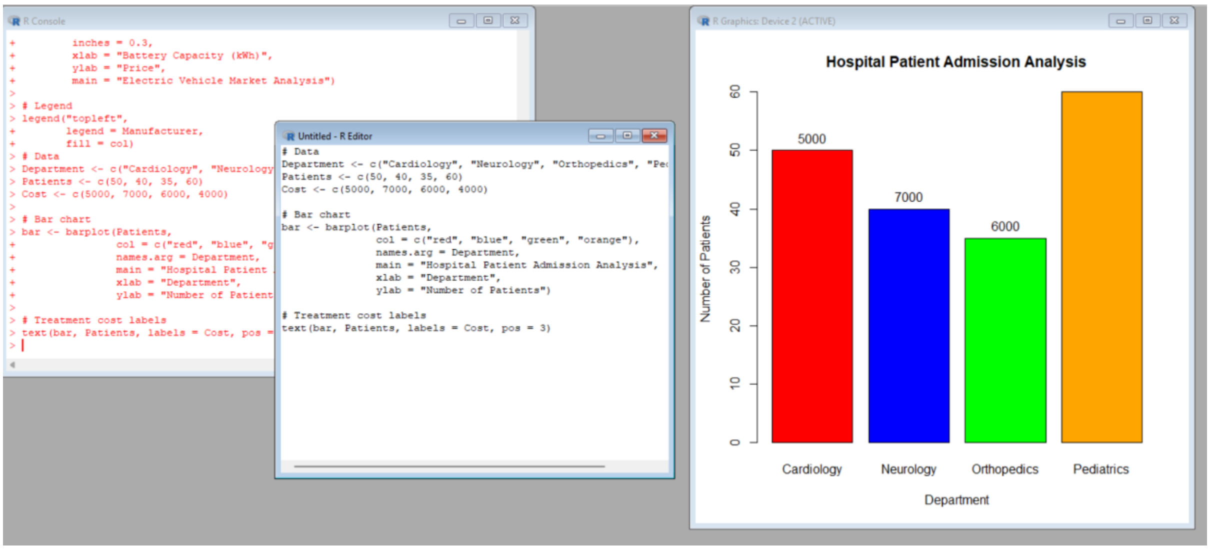


**Scenario 3: Hospital Patient Admission Analysis**

**A. Code**

```
Department <- c("Cardiology", "Neurology", "Orthopedics", "Pediatrics")

Patients <- c(50, 40, 35, 60)

Cost <- c(5000, 7000, 6000, 4000)

bar <- barplot(Patients,

            col = c("red", "blue", "green", "orange"),

            names.arg = Department,

            main = "Hospital Patient Admission Analysis",

            xlab = "Department",

            ylab = "Number of Patients")

text(bar, Patients, labels = Cost, pos = 3)
```

**Output:**

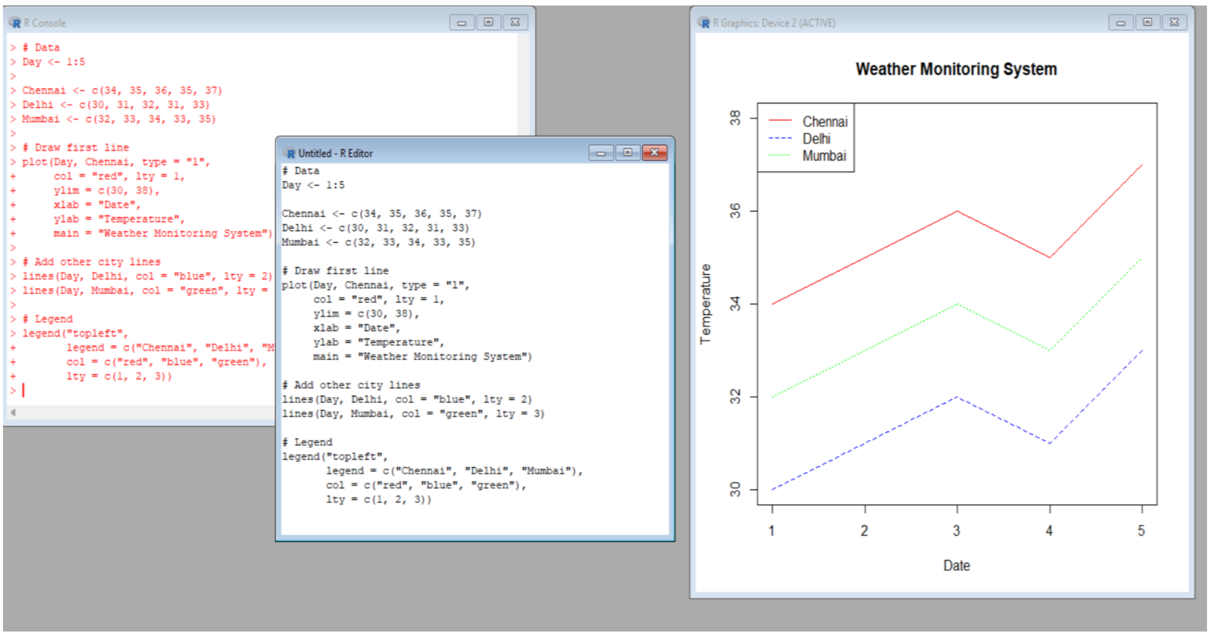


**Scenario 4: Weather Monitoring System**

**A. Code**

```
Day <- 1:5
Chennai <- c(34, 35, 36, 35, 37)
Delhi <- c(30, 31, 32, 31, 33)
Mumbai <- c(32, 33, 34, 33, 35)
plot(Day, Chennai, type = "l",
     col = "red", lty = 1,
     ylim = c(30, 38),
     xlab = "Date",
     ylab = "Temperature",
     main = "Weather Monitoring System")
lines(Day, Delhi, col = "blue", lty = 2)
lines(Day, Mumbai, col = "green", lty = 3)
legend("topleft",
       legend = c("Chennai", "Delhi", "Mumbai"),
       col = c("red", "blue", "green"),
       lty = c(1, 2, 3))
```

**Output:**

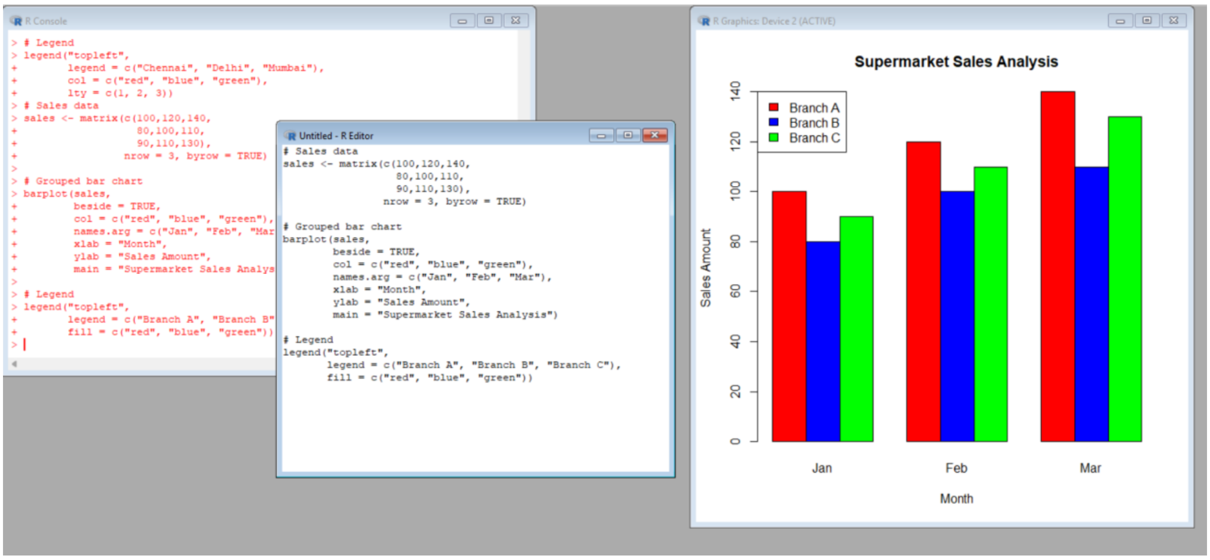


**Scenario 5: Supermarket Sales Analysis**

**A. Code**

```
sales <- matrix(c(100,120,140,
                  80,100,110,
                  90,110,130),
                nrow = 3, byrow = TRUE)
barplot(sales,
        beside = TRUE,
        col = c("red", "blue", "green"),
        names.arg = c("Jan", "Feb", "Mar"),
        xlab = "Month",
        ylab = "Sales Amount",
        main = "Supermarket Sales Analysis")
legend("topleft",
       legend = c("Branch A", "Branch B", "Branch C"),
       fill = c("red", "blue", "green"))
```

**Output:**

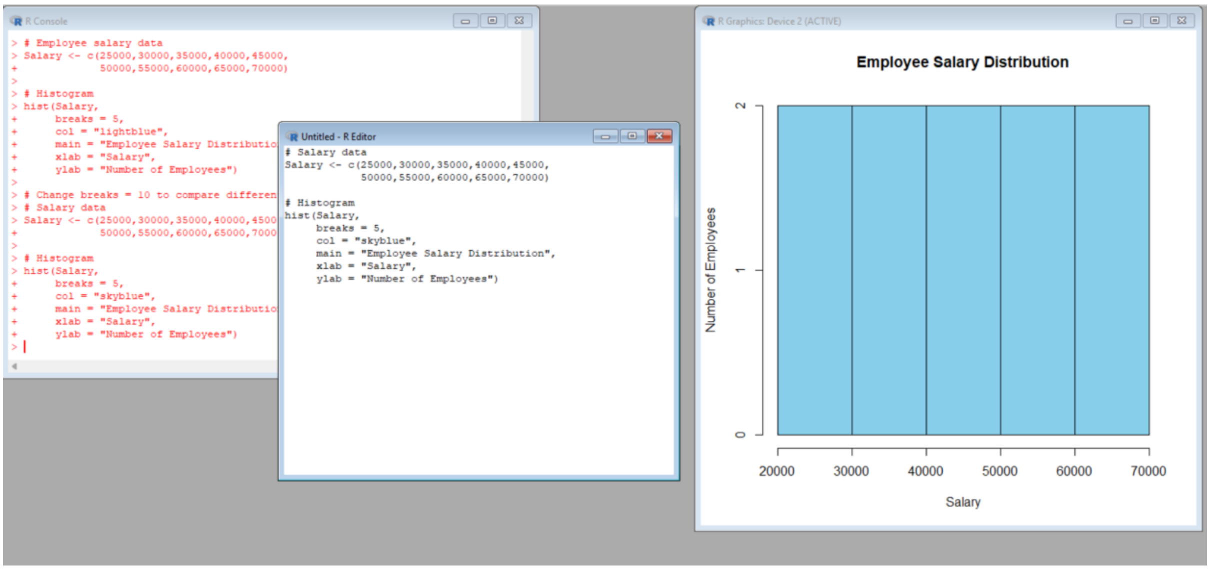


**Scenario 6: Employee Salary Distribution**

**A. Code**

```
Salary <- c(25000,30000,35000,40000,45000,
            50000,55000,60000,65000,70000)
hist(Salary,
     breaks = 5,
     col = "skyblue",
     main = "Employee Salary Distribution",
     xlab = "Salary",
     ylab = "Number of Employees")
```

**Output:**



**Scenario 7: Smartphone Market Comparison**

**A. Code**

```
RAM <- c(4, 6, 8, 12)

Price <- c(15000, 22000, 30000, 45000)

Storage <- c(64, 128, 128, 256)

plot(RAM, Price,

    pch = c(15,16,17,18),

    cex = Storage/64,

    col = c("red","blue","green","orange"),

    main = "Smartphone Market Comparison",

    xlab = "RAM (GB)",

    ylab = "Price")

legend("topleft",

      legend = c("Samsung","Vivo","Realme","OnePlus"),

      pch = c(15,16,17,18),

      col = c("red","blue","green","orange"))
```

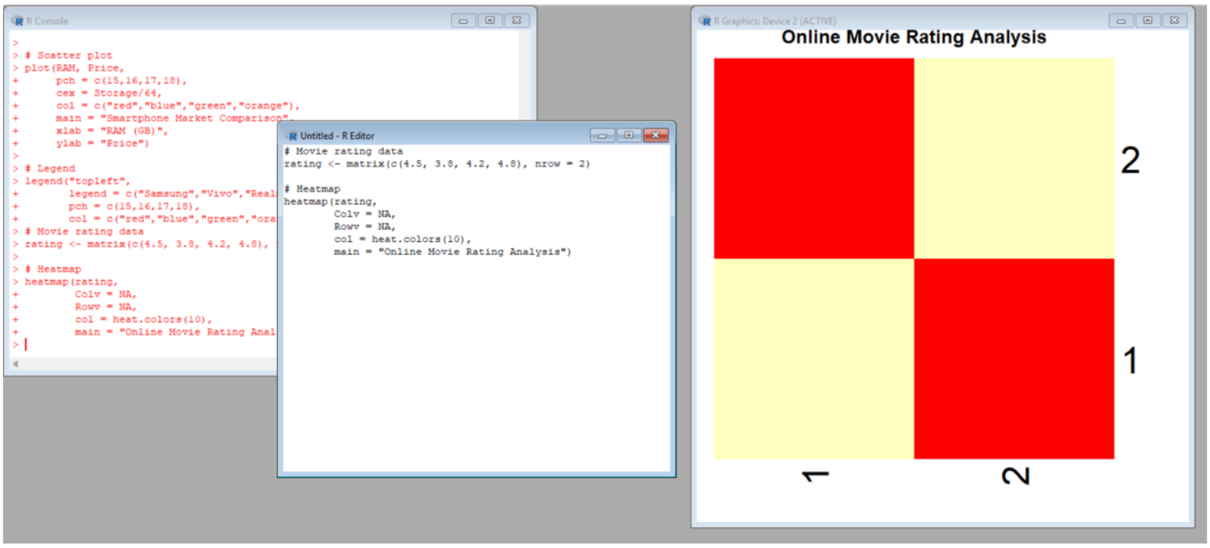
**Output:**

**Scenario 8: Online Movie Rating Analysis**

**A. Code**

```
rating <- matrix(c(4.5, 3.8, 4.2, 4.8), nrow = 2)
heatmap(rating,
        Colv = NA,
        Rowv = NA,
        col = heat.colors(10),
        main = "Online Movie Rating Analysis")
```

**Output:**

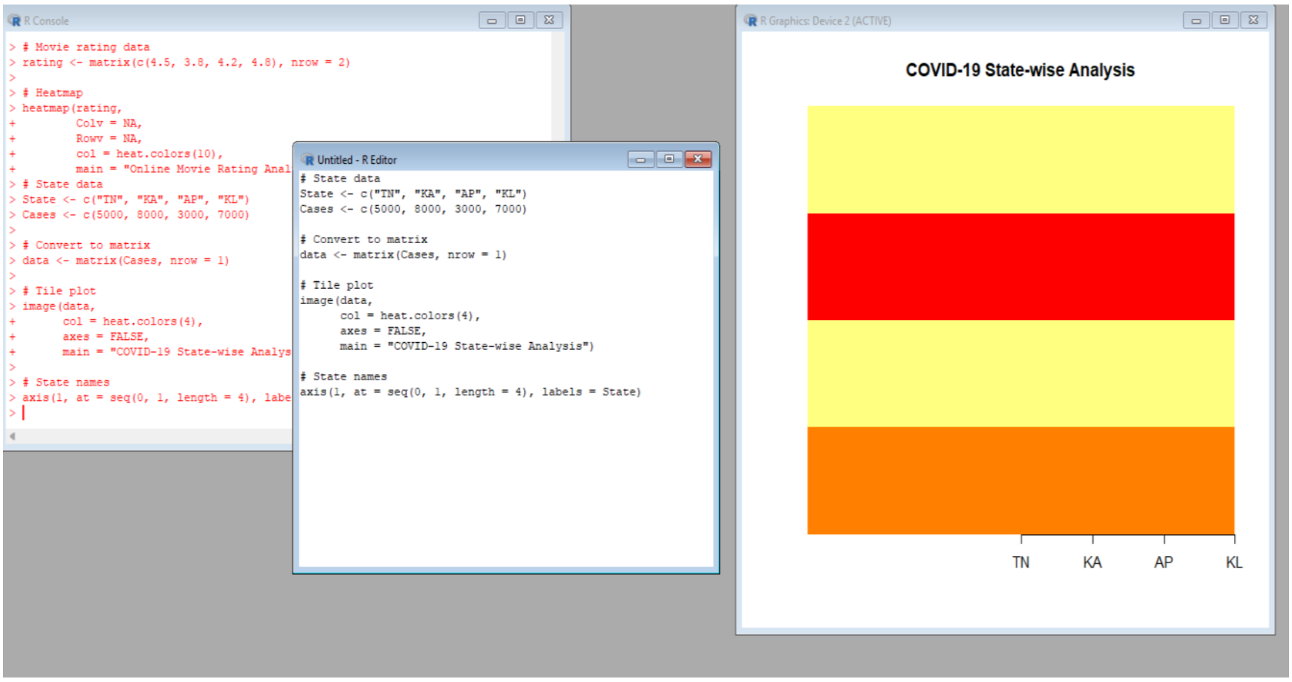


**Scenario 9: COVID-19 State-wise Analysis**

**A. Code**

```
State <- c("TN", "KA", "AP", "KL")

Cases <- c(5000, 8000, 3000, 7000)

data <- matrix(Cases, nrow = 1)

image(data,

      col = heat.colors(4),

      axes = FALSE,

      main = "COVID-19 State-wise Analysis")

axis(1, at = seq(0, 1, length = 4), labels = State)
```

**Output:**



**Scenario 10: Iris Flower Classification**

**A. Code**

```
plot(iris$Sepal.Length,
     iris$Petal.Length,
     col = as.numeric(iris$Species),
     pch = as.numeric(iris$Species),
     cex = iris$Sepal.Width/2,
     xlab = "Sepal Length",
     ylab = "Petal Length",
     main = "Iris Flower Classification")
legend("topleft",
       legend = levels(iris$Species),
       col = 1:3,
       pch = 1:3)
```

**Output:**